

Voice Design TTS의 음성 일관성 분석 및 개선*

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Analyzing and Improving Voice Consistency in Voice Design TTS

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Abstract

Recent advances in text-to-speech (TTS) have been driven by language model (LM)-based architectures, enabling high-quality zero-shot voice cloning through continuation mechanisms where reference audio naturally guides generation. From this foundation, Voice Design TTS has emerged, replacing reference audio with natural language descriptions to enable speech generation without any audio prompt. However, despite this flexibility, Voice Design models lack the continuation-based inductive bias that naturally facilitates consistent voice modeling, leaving generation coherence largely unaddressed. In this paper, we analyze this limitation through a generation-to-generation (Gen-Gen) evaluation protocol and show that Voice Design systems exhibit substantially lower speaker consistency than zero-shot cloning models. To mitigate this gap, we propose a data reconstruction approach at the voice profile level, with a dedicated consistency token to stabilize attention. Our approach improves Gen-Gen cosine similarity from 0.22 to 0.40 in the target setting, corresponding to an 82% relative gain. At the same time, the method reveals a clear trade-off with content fidelity, reflected in higher word error rate (WER). We further analyze the training dynamics behind this trade-off and argue that explicit sink creation can reduce attention efficiency. These findings provide an initial quantitative foundation for studying speaker consistency in Voice Design TTS.

1. Introduction

Recent advances in LM-based TTS have enabled zero-shot voice cloning with remarkable quality [1, 2]. Building on this success, Voice Design TTS has emerged as a complementary paradigm that produces speech directly from natural-language descriptions such as “a calm middle-aged female voice” or “an energetic male narrator”, without requiring reference audio [3, 4]. This setting is attractive for content creation, accessibility tools, and conversational agents as it eliminates the need for reference recordings.

Despite this flexibility, we observe that Voice Design systems often fail to preserve speaker identity across multiple generations conditioned on the same prompt. When the content sentence changes, the generated voice may drift in perceived identity even though the voice description remains fixed. We refer to this phenomenon as *generation-to-generation (Gen-Gen)* inconsistency.

This problem is fundamentally different from the reference-to-generation (Ref-Gen) evaluation protocol used in voice cloning. While Ref-Gen measures how well a synthesized sample matches a given reference recording, Voice Design operates without any canonical reference utterance. The more relevant question is therefore whether the model can repeatedly instantiate the *same* latent speaker identity from the *same* textual voice description.

We argue that the root cause is architectural. In LM-based voice cloning, the model conditions on reference audio tokens and continues the sequence autoregressively. The speaker identity is thus injected through a continuation-style context that remains tightly coupled to generation. By comparison, Voice Design models infer speaker identity from text alone and generate speech from scratch, with no acoustic continuation to stabilize speaker representation.

Based on these observations, our contributions are:

- We analyze the architectural reason why Voice Design TTS is inherently more vulnerable to speaker drift, grounded in the absence of continuation mechanisms.
- We propose a mitigation based on voice profile-level training data reconstruction and attention sink stabilization.
- We analyze the trade-off between voice consistency and attention efficiency via training dynamics behind this approach.

2. Related Work

2.1 Evolution of Zero-shot TTS Architectures

Early zero-shot TTS systems relied on dedicated speaker encoders, conditioning the synthesis model on a single or averaged embedding from one or more reference utterances. While effective for seen speakers, these approaches struggled with truly zero-shot scenarios involving unseen speakers [5].

The LM-based paradigm, pioneered by VALL-E [1], transformed TTS into a conditional language modeling task. By treating speech as discrete codec tokens, VALL-E demonstrated that large-scale language models could achieve zero-shot synthesis through in-context learning. This approach was further developed by subsequent work such as CosyVoice [2], achieving stronger zero-shot speaker similarity at scale.

2.2 Voice Design TTS

Voice Design shifts the goal from cloning an observed speaker to instantiating a speaker from text. Recent systems such as Parler-TTS [3] and Qwen3-TTS [4] have demonstrated that natural-language voice descriptions can serve as controllable prompts for synthesis. This approach is practically attractive because it enables speaker specification without requiring any reference recording.

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However, existing work has focused primarily on achieving target voice characteristics in single utterances, measured through Ref-Gen similarity on annotated test sets. Comparatively little attention has been paid to whether the same description leads to a stable speaker identity across multiple utterances.

2.3 Speaker Consistency Evaluation

Speaker similarity is typically measured with pretrained speaker verification models [6], suited for Ref-Gen comparisons when a target recording is available. For Voice Design, however, consistency must be measured without relying on a fixed reference speaker at inference time. Prior studies have noted the difficulty of maintaining identity in zero-shot cloning [5], yet explicit speaker consistency modeling in Voice Design remains unexplored.

3. Methods

3.1 Problem: The Absence of Continuation Mechanism

LM-based voice cloning models are trained through continuation tasks, where a single utterance is split into a prompt segment and a target segment. The decoder extends this mixed context autoregressively, so speaker identity encoded in the reference audio tokens naturally anchors subsequent generation. Voice Design models, by contrast, receive only a textual voice description with no acoustic prompt (Figure 1A), leaving speaker identity to be inferred from abstract text alone.

Unlike reference audio, which encodes a specific acoustic identity, textual descriptions such as "young female voice" covers a wide region of the acoustic space, leaving the model to sample from an underspecified target at each generation. This underspecification, together with the absence of acoustic continuation, places Voice Design in a structurally similar position to pre-LM voice cloning approaches, which also conditioned synthesis on abstract speaker representations rather than raw audio context and suffered from analogous consistency issues [5]. As shown in Table 1, although Voice Design models remain competitive in naturalness, their Gen-Gen cosine similarity is markedly lower than the voice cloning baseline, indicating severe speaker drift across generations.

3.2 Data Reconstruction for Explicit Voice Modeling

Since Voice Design models cannot leverage continuation mechanisms, they require explicit training to associate voice prompts with consistent speaker identities. We achieve this through a two-step training data reconstruction process.

First, we unify the voice prompt assigned to each speaker group. Each speaker is associated with a fixed speaker-level description, but individual utterances carry varying style attributes. We consolidate them into a single unified voice profile by merging the speaker-level prompt with per-utterance style attributes (e.g., "young female voice" + "energetic" → "young female voice, energetic"), then assign this to every utterance in the group.

Second, we filter out acoustically inconsistent utterances within each voice profile using ECAPA-TDNN embeddings [6], retaining only those whose within-profile similarity exceeds a predefined threshold. After reconstruction, the model is trained to map the same voice prompt consistently to the same speaker identity.

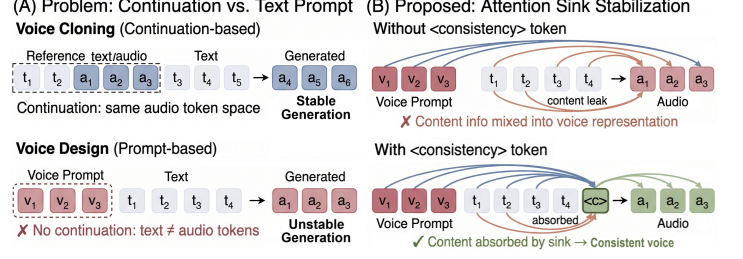


Figure 1: (A) Unlike Voice Cloning, which leverages audio continuation as a natural learning objective, Voice Design relies solely on a textual prompt, making continuation impossible and resulting in an unstable generation process. (B) Without the consistency token, content information leaks into the voice representation, corrupting speaker identity. With the token inserted, it serves as a dedicated information anchor that absorbs content-related attention, yielding a clean and stable voice representation.

Table 1: Evaluation of zero-shot TTS systems.

Model	Task	Params	UTMOS $\uparrow$	WER $\downarrow$	COS _{RG} $\uparrow$	COS _{GG} $\uparrow$
Chatterbox [2]	Cloning	0.5B	3.36	0.16	0.73	0.54
Parler-TTS [3]	Design	0.9B	3.01	0.17	0.41	0.22
Qwen3-TTS-VD [4]	Design	1.7B	3.38	0.14	0.31	0.22

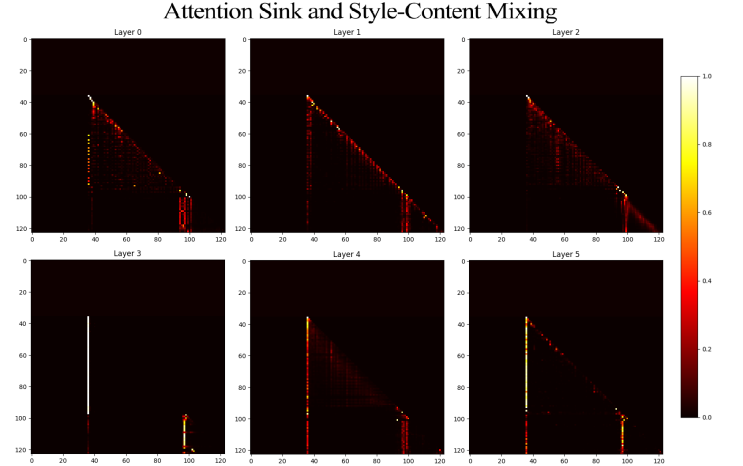


Figure 2: Attention map of the Qwen3-TTS model, revealing multi-sink attention across voice prompt and content token regions.

3.3 Attention Sink Stabilization

We observe a multi-sink pattern in which attention mass concentrates on both the voice description and content token regions (Figure 2). Since the generated audio tokens attend to both regions, speaker representation becomes entangled with content-specific information, causing the perceived speaker identity to shift across utterances even when the voice prompt is fixed.

To address this, we introduce a <consistency> token inserted between the content text and the generated audio sequence (Figure 1B). Under causal attention, this token can attend to all preceding content tokens and serves as a dedicated sink for content information, reducing the need for generated audio tokens to attend directly to content regions. We hypothesize this dedicated sink reduces content leakage into the speaker representation, encouraging the model to derive speaker identity from the voice prompt alone.

Table 2: Effect of the consistency token on zero-shot evaluation.

Model	Architecture	Method	UTMOS $\uparrow$	WER $\downarrow$	COS _{GG} $\uparrow$
Parler-TTS [3]	Enc-Dec	Baseline	3.01	0.17	0.22
		+ Consistency Token	2.91	0.22	0.28
Qwen3-TTS-VD [4]	Dec-Only	Baseline	3.38	0.14	0.22
		+ Consistency Token	3.11	0.34	0.40

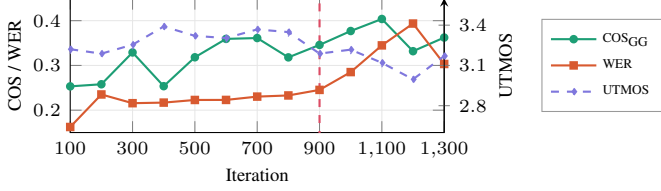


Figure 3: Training dynamics of Qwen3-TTS over iterations, with COS_{GG} and WER on the left axis and UTMOS on the right axis.

4. Experiments

4.1 Experimental Setup

Datasets and Models. We use LibriTTS-P [7] for both training and evaluation. We select two Voice Design architectures with contrasting designs, Parler-TTS (Enc-Dec) and Qwen3-TTS-VD (Dec-Only), to assess generalizability across architectures.

Evaluation Metrics. We use UTMOS for naturalness and WER via Whisper-large-v3 for content fidelity. For voice consistency, we measure COS_{GG}, which computes pairwise ECAPA-TDNN [6] cosine similarity across ten utterances generated with the same voice prompt but different content, repeated over ten prompts. This differs from COS_{RG}, which compares each of 100 generated utterances against its corresponding reference recording.

Table 2 presents the final evaluation results. In both architectures, the proposed method substantially improves COS_{GG} while UTMOS remains stable, with a more pronounced improvement in Qwen3-TTS-VD than in Parler-TTS.

4.2 Training Dynamics Analysis

These results reveal a clear trade-off: our approach improves voice consistency across generations, but this gain is accompanied by a substantial degradation in content fidelity. Figure 3 shows the evolution of key metrics over training iterations. COS_{GG} improves from 0.22 to 0.40 with fluctuations, indicating that the model learns stronger voice-prompt associations over training. At the same time, WER increases gradually at first, then accelerates sharply around iteration 900 (marked by the dashed vertical line as an inflection point) as COS_{GG} breaks through its previous plateau.

These dynamics indicate two regimes. In the early stage, the model improves voice consistency without severely compromising linguistic content. In the later stage, competition for attention becomes stronger, and intelligibility degrades more rapidly. This WER trajectory reflects the operating principle of the consistency token, which addresses two structural issues observed in the attention maps (Figure 2): style-content mixing due to the absence of an explicit anchor in early layers, and dominance of the leading instruct token that blocks style information from reaching audio generation in deeper layers. However, this ultimately introduces a new sink that competes with content modeling for attention capacity, and as it absorbs an increasing share of the attention budget, the model’s ability to represent linguistic content degrades.

5. Discussion and Conclusion

5.1 Limitations and Future Directions

This work has several limitations. First, broader generalization to other Voice Design systems remains to be confirmed. Second, the improvement is more pronounced in the Dec-Only architecture than in the Enc-Dec architecture. We attribute this to Enc-Dec models’ architectural constraint: speaker conditioning is mediated through cross-attention in the encoder, limiting the influence of a decoder-level token on the upstream speaker representation. Third, the observed COS/ WER trade-off remains unresolved. The consistency token may introduce an analogous effect, similar to BOS tokens abandoned in recent LM architectures, creating a sink that competes with linguistic content for attention capacity. Future work may explore self-supervised objectives that directly optimize for cross-utterance speaker invariance, or approaches that align textual voice descriptions with audio representations, recovering the continuation-based inductive bias that exists in Voice Cloning TTS.

5.2 Conclusion

This paper identifies a fundamental problem in Voice Design TTS: unlike LM-based voice cloning, it cannot rely on acoustic continuation to stabilize speaker identity across generations. We propose a training recipe based on voice profile-level data reconstruction and attention sink stabilization. We view this as a first step toward principled consistency modeling in Voice Design TTS, and hope this work motivates further research on stable and intelligible text-driven speech synthesis.

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